

12 Builder Basics · Variables — English Worksheet

Topic: Variables · **Course:** Builder Basics · **Time:** about 45 minutes

1 · Predict 🧠

A **variable** is a name that holds a number. `length = 5` means the name `length` now holds 5.

Read each set of steps. Write what you think will happen.

```
length = 5
repeat length times:
  place block down
  move forward
```

What does the agent build? How long is it?


```
length = 6
repeat length times:
  place block down
  move forward
turn right
repeat length times:
  place block down
  move forward
```

Both sides use the same name. If you change the first line to `length = 9`, how many numbers do you edit, and what happens to both sides?


```
length = 3
length = 8
repeat length times:
    place block down
    move forward
```

How long is the path? Explain what happened to the 3.

2 · Spot the Bug 🐛

Each code block is broken. Read what it should do, rewrite it, then explain why the original was wrong.

Bug A — The agent should build a path **6 blocks long** using a variable.

```
repeat length times:
  place block down
  move forward
length = 6
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Both sides of the corner should be the same length, and one variable should control both.

```
length = 6
repeat length times:
  place block down
  move forward
turn right
repeat 4 times:
  place block down
  move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent should build a path 7 blocks long.

```
length = 7
repeat size times:
    place block down
    move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build a **path or fence** using a variable called **length**. Run it. Then change the value of **length** **one time** and run it again, so you get a second build of a different size.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does. Point at the line that sets your variable.

2 · Your build. Point the camera. Show both sizes you built.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.